



General Chemistry- Chemistry of Elements

Day 01- 17/02/2022

What is Chemistry, and What Can Chemistry Do for You?

You may wonder why

- Why the boiling water bubbles and produces steam while the teakettle retains its original shape?
- How do the tea leaves change the color of the water while the teabag remains as full and plump as ever?
- Why does sugar make your tea sweet, and why is the tea itself bitter?
- How gasoline burns and drives vehicles down the road?
- How does it pollute the air we breathe?
- Why some books that are hundreds of years old are still in good shape while other books that are only 50 years old have pages that are brown, brittle, and crumbling.
- Can the books with damaged pages be saved?



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Chemists can answer all these questions
and
Others like them

Periodic Patterns in the Main-Group Elements

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	72 Hf
7	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	104 Rf

14.1 Hydrogen, the Simplest Atom
Highlights of Hydrogen Chemistry

14.2 Trends Across the Periodic Table:
The Period 2 Elements

14.3 Group 1A(1): The Alkali Metals

14.4 Group 2A(2): The Alkaline Earth
Metals

14.5 Group 3A(13): The Boron Family
Highlights of Boron Chemistry

14.6 Group 4A(14): The Carbon Family
Highlights of Carbon Chemistry

Highlights of Silicon Chemistry

14.7 Group 5A(15): The Nitrogen Family

Highlights of Nitrogen Chemistry

Highlights of Phosphorus Chemistry

14.8 Group 6A(16): The Oxygen Family
Highlights of Oxygen Chemistry

Highlights of Sulfur Chemistry

14.9 Group 7A(17): The Halogens

Highlights of Halogen Chemistry

14.10 Group 8A(18): The Noble Gases

①

Why H doesn't fit much to the group 1A (1)?

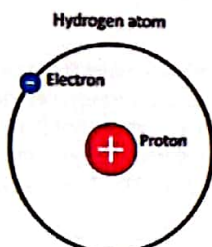
- ① Hydrogen doesn't donate electrons when forming bonds. Instead it shares the only valence electron.
- ② Like a non-metal, H has higher Ionization energy as well as higher electronegativity (relatively to the Li which has ~~the~~ highest IE and EN) in group 1A)

	H	Li
IE (kJ mol^{-1})	1311	520
EN	2.1	1.0

Why H doesn't fit much to the group 1A (1)?

14.1 HYDROGEN, THE SIMPLEST ATOM

Importance of hydrogen



- Nucleus with a single positive charge
- Surrounded by a single electron

- Despite this simple structure, or perhaps because of it, hydrogen may be the most important element of all.
- In the Sun, hydrogen (H) nuclei combine to form helium (He) nuclei in a process that provides nearly all Earth's energy.
- About 90% of all the atoms in the universe are H atoms, making it the most abundant element by far.
- On Earth, only tiny amounts of the free element occur naturally (as H_2), but hydrogen is abundant in combination with oxygen in water.

Properties of H_2 gas

- Colorless
- Odorless
- Very low melting ($-259^\circ C$)
- Boiling points ($-253^\circ C$)

Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

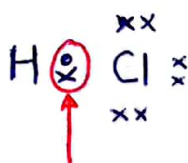
- Some think it fits in Group 1A(1) because of its single valence electron and common 1 oxidation state, However,

- Unlike the alkali metals, hydrogen shares its electron with nonmetals rather than losing it to them.

- Like a nonmetal, it has a relatively high ionization energy (IE 1311 kJ/mol) much higher than that of lithium (IE 520 kJ/mol), whose IE is the highest in Group 1A

- Much higher electronegativity (EN of H 2.1; EN of Li 1.0).

→ As an example let's consider about the HCl (Hydro-Chloric Acid molecule)



Hydrogen atom doesn't give the electron to the Chlorine atom. Instead it shares the electron to electron bond.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

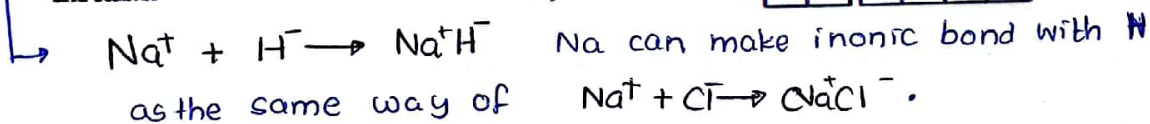
Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Others think hydrogen fits in Group 7A(17) because like the halogens it forms diatomic molecules and fills its outer (valence) level either by electron sharing or by gaining one electron from a metal to form a monatomic anion, the hydride ion (H^- ; O.N. = -1).

- However, hydrogen has a lower electronegativity (EN = 2.1) than any of the halogens (ENs range from 2.2 to 4.0)
- Lacks their three valence electron pairs.
- H^+ ion is rare and reactive, whereas halide ions are common and stable.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								



Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Still others think hydrogen fits best in Group 4A(14) because of a half-filled valence level and similarities in ionization energy, electron affinity, electronegativity, and bond energies. ① ② ③ ④

Hydrogen's unique behavior is attributable to its tiny size.

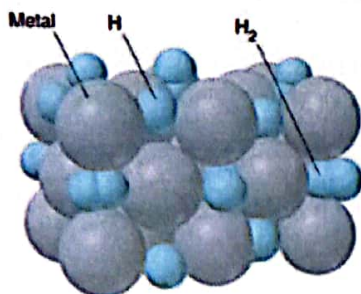
- Hydrogen has a high IE because its electron is very close to the nucleus, with no inner electrons to shield it from the positive charge.
- It has a low EN (for a nonmetal) because it has only one proton to attract bonding electrons.
- In this chapter, hydrogen will appear in either Group 1A(1) or 7A(17) depending on the property being considered.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Highlights of Hydrogen Chemistry

Metallic (Interstitial) Hydrides

- Many transition elements form metallic (interstitial) hydrides, in which H_2 molecules (and H atoms) occupy the holes in the metal's crystal structure. Thus, such hydrides are not compounds but **gas-solid solutions**.



There are spaces between the metallic atoms. Hydrogen molecules or Hydrogen atoms can be trapped in between those spaces.

Interstitial Hydrides

- Also, unlike ionic and covalent hydrides, **interstitial hydrides**, such as $TiH_{1.7}$, typically lack a single, stoichiometric formula because the metal can incorporate variable amounts of hydrogen, depending on the pressure and temperature of the gas.

The ratio between the number of atoms of Titanium and Hydrogen can be changed due to the conditions of the environment.

(pressure and temperature etc.) In a higher pressure and a higher temperature more H atoms or molecules can be trapped in the spaces.

14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS



- Electrons fill the one ns and the three np orbitals according to Pauli's exclusion principle and Hund's rule.
- As a result of increasing nuclear charge and the addition of electrons to orbitals of the same energy level (same n value), atomic size generally decreases, whereas first ionization energy and electronegativity generally increase (see bar graphs on p. 557).
- Metallic character decreases with increasing nuclear charge as period members change from metals to metalloids to nonmetals.
- Reactivity is highest at the left and right ends of the period, except for the inert noble gas.
- Bonding between atoms of an element changes from metallic, to covalent in networks, to covalent in individual molecules, to none (noble gases exist as separate atoms). As expected, physical properties, such as melting point, change abruptly at the network/molecule boundary, which occurs in Period 2 between carbon (solid) and nitrogen (gas).
- Bonding between each element and an active nonmetal changes from ionic, to polar covalent, to covalent. Bonding between each element and an active metal changes from metallic to polar covalent to ionic.

When the number of protons in the nucleus increases, the power of the nucleus goes higher. So the electrons can move closer to the nucleus. Because of that the size of the atom decreases.

As a result metallic structure also decreases.

$$\begin{array}{r} 9 \overline{) 30.5} \\ \underline{27} \\ 35 \\ \underline{36} \\ -5 \end{array}$$

14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS

- The acid-base behavior of the common element oxide in water changes from basic to amphoteric to acidic as the bond between the element and oxygen becomes more covalent.
- Reducing strength decreases through the metals, and oxidizing strength increases through the nonmetals. In Period 2,
 - Common oxidation numbers (o.N.s) equal the A-group number for Li and Be
 - A-group number minus eight for O and F
 - Boron has several o.N.s
 - Ne has none, and
 - C and N show all possible o.N.s for their groups

when the oxides dissolve in water.

Oxide

Basic $\rightarrow$ Amphoteric $\rightarrow$ Acidic

$\xrightarrow{\hspace{2cm}}$
A-X bond
covalency $\uparrow$

(LO $\rightarrow$ NO)